How accurate LIGHT_Kr v0_4 is

The pure-krypton model LIGHT_Kr v0_4, version 0.4.0-kr, on 15 October 2026: v0_3 with its four 5s levels resolved, each forming the excimer at its own measured rate, the excimer and the molecular ion redissociating by detailed balance, and the two-temperature closure's tail made for a field-driven discharge, scored against pure-krypton measurements, with its propagated uncertainty, its reference point and its checks

Bottom line. No measurement in pure krypton covers v0_4's own regime, a gas at 0.1 to 2 atm whose electrons a field heats. The measurements that exist test its parts. Only electron swarm measurements train anything, the cross sections, which v0_4 takes unchanged from v0_3; every other measurement only scores the model.

- **What v0_4 changes.** v0_3 lumped the four 5s levels into one Kr^* that formed the excimer Kr_2^* at $1.7 \times 10^{-32}\,cm^6s^{-1}$, a beam experiment's value [19]. v0_4 carries each level at its own energy, and each forms Kr_2^* at its own measured rate: the metastable $1s_5$ ($^3P_2$) at $3.8 \times 10^{-32}\,cm^6s^{-1}$ and the resonant $1s_4$ ($^3P_1$) at $9.2 \times 10^{-32}\,cm^6s^{-1}$, both from selective excitation in pure krypton [56]. Each level is pumped to the 5p rung on its own measured cross sections [10], fed by the 5p lines on their measured branching [70], and the levels exchange population on krypton at their measured rates [59, 58]. The excimer and the ion Kr_2^+ now redissociate in hot gas, by detailed balance on their measured binding energies [60, 61, 62].
- **The answer moves by a factor of 2.2.** At LIGHT's reference point (1 atm, $300\,kW\,cm^{-3}$ for 1 μs) v0_4 sends 31.22 % of the input to 147 nm on the kinetic EEDF; v0_3 sent 14.05 %. With every level forming the excimer at v0_3's rate it would send 11.92 %: a faster formation lets more Kr^* escape the electrons' 5s $\leftrightarrow$ 5p cycle, which still turns 57.0 % of the input into heat. Over the run the excimer's redissociation undoes 5.8 % of its formations, as the gas heats to 707 K.
- **Propagated from its sources**, the share carries $\pm$20.1 points in quadrature ($\pm$64 %), and the trained cross sections' posterior $\pm$0.11. The largest term is the pumping of the two $J = 1$ levels, whose 5s $\rightarrow$ 5p cross sections nothing measures; with the unmeasured Kr_2^* + Kr_2^* and stepwise ionization, three of the four largest terms have no measurement behind them (Section 1).
- **A new score that nothing trained on.** Mano et al.'s krypton electroluminescence yield [67], the light per electron and per centimetre of drift at 1.1 bar, tests the excitation cross sections in the swarm regime. v0_4's swarm, each excitation weighted by the photons the model's own cascade makes of it, lies at 0.99 to 1.04 of their fit from 6 to 13.5 Td (Section 6).
- **The electron swarm** is v0_3's: on the train sets the drift velocity is within 2.8 % rms and the ionization coefficient within 9.9 % from 50 to 300 Td, scored as the experiments measure them; below 50 Td the measured coefficient lies -36 % from the model on average, where the measurements part by up to $\times$4.0 (Section 5).
- **Under an electron beam** v0_4 turns 0.543 of the deposited energy into 147-nm light at low pump, where Morozov et al. measured 0.42 $\pm$ 0.05 [39]: an error of +29 % (v0_3: +29 %). At $3 \times 10^5\,W\,cm^{-3}$ and 1 atm it is +1 % from Eckstrom et al.'s report (v0_3: -26 %).
- **The two-temperature closure** reads a tail table that v0_4 builds from its own kinetic EEDF for a field-driven discharge. At the reference point it gives 32.92 %, +5 % from the kinetic EEDF; v0_3's table, made for beam-driven Kr/F_2 states, gives +8 %. Over the scan's corners the closure stays within 5 % of the kinetic EEDF (Section 4).
- **The numerical error is far smaller.** The energy ledger closes to 7×10^{-15} of the input; halving the kinetic closure's renewal steps or doubling its cells moves the 147-nm and near-IR shares by at most 2×10^{-5} of themselves, and time-stepping the afterglow's EEDF, rather than holding its steady shape, by 2×10^{-5}.

Table 1 lists every test, Section 1 reads them as errors and propagates the model's own uncertainty, and the sections after it give each test's detail. The swarm datasets in the role *train* set the cross sections' factors (Section 5); every other measurement in the table only scores the model, and no measurement

set any other rate, which come from the sources in Appendix A.

test	measured	v0_3's error	v0_4's error	role
energy ledger closure, reference point			7×10^{-15} of E_{dep}	bookkeeping
energy balance at every node of the flow network			25 nodes, to 10^{-8}	bookkeeping
drift velocity, the 3 sets that agree [41, 43, 34], 79 points		rms 2.8 % (one set of cross sections)		train
ionization coefficient, 50 to 300 Td, the 5 sets that agree [50, 44, 5, 36, 37], 64 points		rms 9.9 %		train
drift velocity, the 8 sets that do not agree (1953 to 1992)		rms 8 to 47 % by set		held out; the two at 195 K validation
characteristic energy D_T/μ, Koizumi et al. [48], 26 points, 0.005 to 0.9 Td		rms 7.1 %		validation
ionization coefficient below 50 Td, 5 sets [51, 36, 37, 44, 5], 36 points		mean -36 %		held out by rule
electroluminescence yield, 6 to 13.5 Td, Mano et al. [67]	$0.113\,E/N - 0.3$		0.99 to 1.04 of it	new, held out
ionization coefficient, Monte Carlo on the same cross sections [3], 20 to 50 Td		Biagi's file +3.7, -4.0, -3.5, -2.5 %		solver check
the Monte Carlo's bulk drift velocity and SST coefficient, trained set against Biagi's file, 30 to 300 Td		within 1.6 %		solver check
energy per ion pair under a beam, from Saito et al. [38]	23.9 eV	+10 %		held out
excitations per ion pair, Saito et al. [38]	0.55	+24 %		held out
energy per excitation or ion, Saito et al. [38]	15.4 eV	+1.0 %		the beam's check
VUV efficiency, continuous e-beam, Morozov et al. 2008 [39]	0.42 ± 0.05	+29 %	+29 %	held out
VUV efficiency at 3×10^5 W cm^{-3}, Eckstrom et al. 1988 [40]	near 0.46, their limit	-26 %	+1 % at 1 atm	held out

Table 1: LIGHT_Kr v0_4 against the data, every band at its default, on the kinetic EEDF, beside v0_3's errors. v0_4 runs on v0_3's trained cross sections, so the swarm and the beam's partition score one set of cross sections. An error is the model less the measurement over the measurement, signed for a single value and rms (or a mean) over a set. A validation set is held out of the training but took part in choosing which factors train (Table 10). The swarm rows compare the model as each experiment measures it: the two-term solution times the Monte Carlo's ratios (Section 5). The beam rows run under the e-beam drive of Section 7.

1 Overall accuracy and its uncertainty

Against the data. Four groups of measurements bear on v0_4, and each tests a part of it (Table 1).

- **The electron swarm** (Section 5) trains the cross sections that give every electron rate and scores what the training did not see. v0_4 runs on v0_3's trained set. On the 8 sets that agree with one another the drift velocity is within 2.8 % rms and the ionization coefficient within 9.9 % from 50 to 300 Td. Held out, the characteristic energy of Koizumi et al. is within 7.1 % rms (Biagi's file: 15.8 %), a validation rather than a test, as it took part in choosing which factors train. The ionization coefficient below 50 Td, which nothing in the training saw, lies -36 % from the measurements on average, where those part by up to $\times 4.0$ from one another.
- **The electroluminescence yield** (Section 6) is new in v0_4: a swarm measurement of the light the

excitation makes, which no fit saw. The model lies at 0.99 to 1.04 of it from 6 to 13.5 Td.

- **The e-beam measurements** (Section 7) score the chain from excitation to the 147-nm photon. The beam's partition is v0_3's: +10 % from Saito et al. on the energy per ion pair and +24 % on the excitations per pair. At low pump the error against Morozov et al. is +29 %, 2.5 of their standard errors. At 3×10^5 W cm^{-3} and 1 atm it is +1 % against Eckstrom et al.'s report, and it loses 14 % below its own ceiling.
- **Nothing scores the field-driven cycle.** No pure-krypton measurement in hand tests the 5s $\leftrightarrow$ 5p cycle under a field, which with the excimer's formation decides v0_4's answer. Its cross sections are measured, and their error is propagated below.

Propagated from the sources. Each input is set to the two ends its source allows and the run repeated, the others at their defaults, on both electron closures (Table 2). The inputs are the 17 bands the rows read and the 3 measured constants with a quoted error that no band covers: $Kr_2^+ + e^-$ ($\pm$10 % [30]), $Kr^+ + 2Kr$ (the spread of the three measurements [26, 27, 28]) and the Kr_2^* lifetime (270 $\pm$ 5 ns [25]). Half the difference between an input's two ends is its term. The trained cross sections are one more input: the reference point is run at 12 draws of the training's posterior on the kinetic EEDF and 40 on the two-temperature closure, and the draws' standard deviation is their term. The quadrature sum treats the terms as independent standard deviations, which a band's range is not, so it is indicative. The excursions from the reference, summed by sign, are the ceiling of the spread, not a bound. Three kinds of input are not propagated: the 5p rung's lifetime and its branching (its light is under 1 % of the input), the transfers between the 5s levels (no error is quoted; Table 3 gives the share without them), and the excimer's rotational constant, an estimate that scales its redissociation as $1/r_e^2$ while the well's band moves it exponentially.

On the 147-nm share at the reference point, 31.22 % of the input on the kinetic EEDF, the quadrature sum is $\pm$20.1 points ($\pm$64 %). The excursions sum to $-$73.5 and +46.3 points. The largest terms are the $J = 1$ levels' pumping (11.79 points), $Kr_2^* + Kr_2^*$ (9.66) and $e^- + Kr^* \rightarrow Kr^+$ (7.30). The trained cross sections' term is 0.11 points (0.4 % of the share). On the two-temperature closure the sum is $\pm$20.6 points ($\pm$63 %), the cross sections' term 0.20.

What dominates has changed since v0_3, where the excimer's formation did. Three of the four largest terms are rates nothing measures. The two $J = 1$ levels' 5s $\rightarrow$ 5p cross sections are their same-core metastables' as a proxy; with no electron pumping them the share rises to 54.81 %, as the resonance levels, which feed the fast $1s_4$ formation, then escape the cycle. $Kr_2^* + Kr_2^*$ at its band's top cuts the share by 62 %, and the stepwise ionization of the 5s levels at its top by 47 %. Of the measured inputs, the $1s_4$ formation's band, down to a third of the $1s_5$ rate (Audouard et al.'s ratio [57]), takes the share to 18.44 %, and the measured 5s $\rightarrow$ 5p cross sections' quoted errors span 27.37 to 37.52 %. The bands that are off by default though a measurement bounds them move it least: the excimer's quenching at its top by $-$6 %, the fast state's share by 0 %, the recombination's gas-temperature factor by +8 % and the resonance lines by 0 %. The Hornbeck–Molnar share at its top moves it by $-$0.12 %: direct and associative ionization make a small part of the ions next to Penning's.

input	its two ends	147 nm, kinetic	147 nm, two-temperature	near IR, kinetic	T_g (K), kinetic
$e^- + Kr^*(J=1) \rightarrow Kr^{**}$, scale	0 to 1	54.81 to 31.22	56.20 to 32.92	0.12 to 0.47	566 to 707
$Kr_2^* + Kr_2^*$, Penning	0 to 5.0×10^{-10} cm^3 s^{-1}	31.22 to 11.90	32.92 to 12.42	0.47 to 0.62	707 to 821
$e^- + Kr^* \rightarrow Kr^+ + 2e^-$	0 to 1.5×10^{-7} cm^3 s^{-1}	31.22 to 16.63	32.92 to 17.86	0.47 to 0.53	707 to 793
$Kr^*(1s_4) + 2Kr \rightarrow Kr_2^* + Kr$	1.3×10^{-32} to 9.2×10^{-32} cm^6 s^{-1}	18.44 to 31.22	19.39 to 32.92	0.59 to 0.47	781 to 707
$e^- + Kr^* \rightarrow Kr^{**}$, scale	0.6 to 1.4	37.52 to 27.37	39.49 to 28.82	0.40 to 0.52	669 to 729
$Kr^* + Kr^*$, Penning	1.5×10^{-10} to 5.0×10^{-10} cm^3 s^{-1}	38.84 to 31.22	40.68 to 32.92	0.41 to 0.47	661 to 707
$e^- + Kr^{**} \rightarrow Kr^+ + 2e^-$	0 to 2.0×10^{-7} cm^3 s^{-1}	31.22 to 23.62	32.92 to 25.52	0.47 to 0.52	707 to 752
$Kr^*(1s_5) + 2Kr \rightarrow Kr_2^* + Kr$	1.7×10^{-32} to 4.2×10^{-32} cm^6 s^{-1}	25.69 to 32.16	27.06 to 33.91	0.52 to 0.46	739 to 701
its products over the 5s levels, share	0 to 1	27.52 to 31.22	29.09 to 32.92	0.42 to 0.47	729 to 707
$Kr^*(1s_3) + 2Kr \rightarrow Kr_2^* + Kr$	0 to 9.2×10^{-32} cm^6 s^{-1}	29.53 to 33.18	31.18 to 34.90	0.49 to 0.45	716 to 695
$Kr_2^+ + e^-$, the gas-temperature factor	0 to 1	31.22 to 33.73	32.92 to 35.39	0.47 to 0.45	707 to 692
$Kr^{**} + Kr \rightarrow Kr^* + Kr$	3.0×10^{-11} to 1.7×10^{-10} cm^3 s^{-1}	33.64 to 31.22	35.23 to 32.92	2.44 to 0.47	680 to 707
the excimer's well (its redissociation)	0.62 to 0.709 eV	29.32 to 31.22	31.16 to 32.92	0.49 to 0.47	717 to 707
$Kr_2^* + Kr \rightarrow 3Kr$	0 to 8.6×10^{-15} cm^3 s^{-1}	31.22 to 29.49	32.92 to 31.09	0.47 to 0.47	707 to 717
$Kr_2^+ + e^- \rightarrow Kr^{**} + Kr$	± 10 %	30.76 to 31.64	32.45 to 33.35	0.48 to 0.47	709 to 704
$Kr^+ + 2Kr \rightarrow Kr_2^+ + Kr$	2.2×10^{-31} to 3.3×10^{-31} cm^6 s^{-1}	31.10 to 31.60	32.78 to 33.35	0.47 to 0.47	707 to 704
the electron cross sections (trained)	12 and 40 draws of the posterior	31.28 ± 0.11	32.94 ± 0.20	0.47 ± 0.00	706 ± 1
the $1s_4$ formation's share into 0_u^+	0 to 0.17	31.22 to 31.30	32.92 to 32.99	0.47 to 0.47	707 to 706
Kr_2^* lifetime	265 to 275 ns	31.19 to 31.25	32.89 to 32.95	0.47 to 0.47	707 to 707
$Kr(nl) + Kr \rightarrow Kr_2^+ + e^-$, the share	0 to 1	31.22 to 31.18	32.92 to 32.91	0.47 to 0.47	707 to 707
the resonance lines' escape	0 to 1.7×10^5 s^{-1}	31.22 to 31.19	32.92 to 32.88	0.47 to 0.47	707 to 706

Table 2: The propagated uncertainty's inputs, each alone at its two ends with the others at their defaults, and the outputs at those ends (low end to high end), percent of E_{dep}; for the trained cross sections the mean and standard deviation over the posterior's draws. With every input at its default the 147-nm share is 31.22 % (kinetic) and 32.92 % (two-temperature).

Numerical accuracy.

- **The ledger.** The energy ledger closes to 7×10^{-15} of the input on the kinetic EEDF and to 1×10^{-15} on the two-temperature closure. The energy flows also balance at each of 25 nodes to 10^{-8}: the electron gas, each reservoir, each group of rows and each output.
- **The kinetic EEDF at the reference point.** The run takes 348 kinetic solutions, 274 in the pulse, at E/N from 1.34 to 17.5 Td. Over the pulse's second half the field carries 1.0004 to 1.0007 of the input. There the kinetic and the model's energy balances agree to 7×10^{-6} of the input, and the kinetic mean energy equals the model's to 2×10^{-11}.
- **Convergence.** Half the renewal step moves the 147-nm share by 2×10^{-5} of itself. A renewal at least every 5 ns moves it by 2×10^{-5}, and twice the energy cells by 3×10^{-6}. A ten times tighter solver tolerance moves it by 2×10^{-8}, the EEDF renewed 348 times against the reference's 348.
- **The afterglow's EEDF.** Through the pulse the kinetic EEDF is advanced in time from its last solution; after the pulse it takes its steady shape at each renewal. Advancing it in time after the pulse too (74 time steps) moves the 147-nm share by 2×10^{-5} of itself, the near IR by 1×10^{-5} and the reaction heat by 9×10^{-6}: the steady shape costs nothing at this point.
- **Restarts.** Where the kinetic EEDF finds no solution from its warm start, the solver starts it again cold. The mean match has one solution from any start, so a restart changes no answer, only the way to it. Over the 63 kinetic runs of this note the solver restarted 11 times, in one run. Where a table point told to leave the growth term out has no steady solution, as in the afterglow of very hot gas, where Penning ionization makes electrons faster than recombination takes them, it takes the growth term on the same grid, and failing that a grid of its own: 6 times, in one run. Before v0_4 such a point stopped the run.
- **The swarm scores** are independent of the grid: half and twice the 1000 energy cells move the train sets' rms by 0.01 points at most. The emulator the training ran on reproduces the solver to 0.10 % in the drift velocity and 0.74 % in the ionization coefficient on designs it was not trained on.

2 What v0_4 is

LIGHT_Kr v0_4 is a 0-D model: one uniform box of pure krypton at constant volume, into whose electrons a given power is put. It follows where that energy goes: into the four 5s levels (Kr^*: $1s_5$, $1s_4$, $1s_3$, $1s_2$) and the 5p rung (Kr^{**}), the ions Kr^+ and Kr_2^+, the excimer Kr_2^* in its slow (1_u, 0_u^-) and fast (0_u^+) states

and its 147-nm second continuum, the near-infrared 5p $\rightarrow$ 5s lines, the resonance lines, heat, and the electron gas. Its energy ledger closes by construction. Every species carries a formation energy, and every row's energy goes to the electrons, the gas, light or storage.

Its network has 126 rows on 10 species (Appendix A):

- electron-impact excitation of each of the 51 levels of Biagi's cross sections at its own threshold, ionization and elastic loss [1, 6, 7], with v0_3's factors trained on the swarm data (Section 5): each 5s level makes its own species, every higher level the 5p rung Kr^{**} (the higher levels cascading through it, the excess heating the gas);
- the Hornbeck–Molnar ionization of the 20 levels at or above Kr_2^+'s energy [4], a band at 0;
- Penning ionization of the 5s levels on one another;
- ion conversion $Kr^+ \rightarrow Kr_2^+$ and its reverse, and dissociative recombination of Kr_2^+ onto the 5p rung;
- three-body excimer formation out of each 5s level at its own rate, its reverse, the 147-nm emission of the two excimer states, and their quenching by krypton (a band at 0);
- the transfers between the 5s levels on krypton and their reverses, and the escaping resonance lines (a band at 0);
- the rung's relaxation on krypton and its near-infrared radiation, both branching over the 5s levels;
- the electrons' 5s $\leftrightarrow$ 5p cycle out of each 5s level on Jung et al.'s measured cross sections [9, 10];
- and, as bands at 0, the rows with no measurement.

The electrons' rates come from the two-term Boltzmann equation with electron-electron collisions [8], solved during the run at the model's own mean energy (the default, minutes a run). The alternative is a Maxwellian with a tabulated tail (the two-temperature closure, seconds a run; Section 4). A rate known only as a published spread, or not at all, is a band: a default and two ends.

What changed from v0_3. v0_3 had 85 rows; its Kr^* was the four 5s levels as one species at the $1s_5$ energy, which formed Kr_2^* at Ueda et al.'s $1.7 \times 10^{-32}\,cm^6s^{-1}$ [19].

- **The 5s levels, resolved.** Each level is its own species at Biagi's threshold, made by its own G1 row. The excimer forms out of $1s_5$ at $3.8 \times 10^{-32}\,cm^6s^{-1}$ and out of $1s_4$ at $9.2 \times 10^{-32}\,cm^6s^{-1}$, Marchal et al.'s two values by selective excitation in pure krypton [56]; out of $1s_3$, whose value is not in a source in hand, at the $1s_5$ rate as a proxy. The $1s_5$ value lies with Turner's 4×10^{-32} and Smith and Turner's 4.1 to 4.2×10^{-32} from discharge afterglows [59, 22]; Tracy and Oskam's 2.59×10^{-32} [21] and Ueda et al.'s, from an e-beam at 0.5 to 1 atm, lie below, the latter at the band's low end. $1s_2$ forms the excimer through $1s_4$: above $4 \times 10^{20}\,cm^{-3}$ its decay runs fully through $^3P_1$, by a three-body transfer of about $0.2 \times 10^{-32}\,cm^6s^{-1}$ [58]. $1s_4$ relaxes to $1s_5$ on krypton at Turner's $1.3 \times 10^{-13}\,cm^3s^{-1}$ at 295 K, through his 196-K point [59]; its reverse by detailed balance matches his measured $1s_5 \rightarrow 1s_4$, $7.6 \times 10^{-16}\,cm^3s^{-1}$, to 2 %.
- **The 5s $\leftrightarrow$ 5p cycle, level by level.** Each level is pumped on its own measured cross sections (the two metastables on Jung et al.'s sets, the two $J = 1$ levels on their same-core metastable's, as v0_3's lump took them), and its reverse is detailed balance over its own gap with its own statistical weight. The four rates average to v0_3's lumped one by weight, to 2×10^{-8}. The 5p lines branch over the 5s levels as NIST's Einstein coefficients weighted by $2J + 1$ [70], and Kr^{**} + Kr's products by the levels' weights (band `kps_5s`).
- **Redissociation by detailed balance.** $Kr_2^* + Kr \rightarrow Kr^* + 2Kr$ is the formation rate over the equilibrium constant of the excimer on its measured well, $D_0 = 0.709$ eV (the A state's D_e less half its low-v quantum [60, 61]; band to Killeen and Eden's 0.62 eV), and $Kr_2^+ + Kr \rightarrow Kr^+ + 2Kr$ likewise on the ion's measured 1.149 eV [62]. At the fill's density the excimer's bound share is 27 times its atoms' at 700 K and falls to 1 at 988 K.
- **Bands for what is bounded but not measured at the fill.** The fast 0_u^+ excimer, made from $1s_4$ (0 to Audouard et al.'s share above $6 \times 10^{19}\,cm^{-3}$ [57]); Kr_2^* quenching by krypton (0, as Oka et al. saw no collisional step [65], to Firestone and Selander's measurement and its error [64]); a gas-temperature factor on $Kr_2^+ + e^-$ (0 to Cunningham and Hobson's shock-tube coefficients at $T_e = T_g$ over Shiu and Biondi's law [63, 30]); the escaping resonance lines (0 in a box with no size, to the pressure-independent term of Janssens et al.'s $^3P_1$ loss rate [66]).

The cross sections, the swarm, Jung et al.'s tables and every row that books no 5s level and no excimer are v0_3's. Table 3 undoes each change alone at the reference point.

undone from v0_4	147 nm, kinetic	change	two-temperature	T_g end (K)
none (v0_4)	31.22		32.92	707
every level forms the excimer at Ueda's 1.7×10^{-32} (v0_3's)	11.92	−62	12.58	819
$1s_4$ forms it at the $1s_5$ rate	23.05	−26	24.29	755
Kr^{**} + Kr all to $1s_5$	27.52	−12	29.09	729
no redissociation of Kr_2^*	31.68	+1	33.33	704
no redissociation of Kr_2^+	31.22	0	32.92	707
no transfer between the 5s levels	31.17	0	32.87	707
the $J = 1$ levels pumped by no electron	54.81	+76	56.20	566
v0_3	14.05		15.78	808

Table 3: What each of v0_4's changes does at the reference point: each undone alone, the others kept, percent of E_{dep}; change is relative to v0_4 on the kinetic EEDF. v0_3's lumped Kr^* is the last row.

The formation rates are most of the change. With every level forming the excimer at v0_3's rate the share is 11.92 %, below v0_3's 14.05 %: the rest of v0_4 alone would lower it. $1s_4$'s own rate, against the $1s_5$ rate, is worth the difference from 23.05 %. The rung's relaxation onto every level by weight rather than onto $1s_5$ alone is worth the difference from 27.52 %, as more of it lands on $1s_4$. At the reference point's 707 K the excimer's redissociation costs little (31.68 % without it), the ion's nothing (31.22 %), and the transfers between the levels little (31.17 %). The pumping of the two $J = 1$ levels moves it most: without it the share is 54.81 %. v0_3's lumped cross section carried their proxies too, in the same proportion.

3 The reference point

LIGHT's drive in pure krypton: 1 atm at 300 K, 300 kW cm^{-3} for 1 μs into the electrons, a 10^{15} cm^{-3} seed, 2 μs in all. Table 4 gives its outputs on both closures.

	kinetic EEDF (default)	two-temperature
147 nm, Kr_2^*	31.22	32.92
near IR, 5p → 5s	0.47	0.45
heat, reactions and relaxation	65.45	63.16
heat, elastic collisions	3.21	3.81
stored in ions (the seed's ions recombine)	-0.74	-0.74
to Kr^* and Kr^{**} by electron impact, gross	41.02	43.01
to Kr^+	5.01	5.37
to Kr_2^* (formed × 8.43 eV)	33.45	34.96
Kr^* formed that the electrons pump to 5p (%)	91.8	91.1
Kr^* formed that becomes Kr_2^* (%)	6.9	7.5
Kr_2^* formed out of $1s_5$, $1s_4$, $1s_3$ (%)	41.2, 52.6, 6.2	40.9, 53.0, 6.1
Kr_2^* formed that redissociates (%)	5.8	5.0
Kr_2^* formed that radiates (%)	93.3	94.2
5s → 5p pumpings per Kr_2^* formed, net	14	13
T_e, pulse mean (eV)	2.55	2.65
T_g at the end (K)	707	697
n_e at the end (cm^{-3})	8.2×10^{12}	8.2×10^{12}

Table 4: The reference point at $t = 2$ μs, percent of E_{dep} unless a row says otherwise. The fate rows (the first five, with the other stores) sum to 100; the flows below them overlap.

The electrons pump Kr^* to the 5p rung faster than three-body collisions turn it into Kr_2^*. Krypton relaxes

the rung back, and each level's gap, 0.660 to 1.388 eV, becomes heat: 57.0 % of the input, 15 eV for each 147-nm photon. Of the pumping, 41.9 % is out of $1s_5$, 24.6 % out of $1s_4$, 25.5 % out of $1s_2$ and 8.0 % out of $1s_3$. The rung relaxes onto every level by its weight and radiates onto each by its branching, so the $J = 1$ levels, which the electrons pump on their metastables' cross sections, are refilled from it, and $1s_4$ forms 52.6 % of the excimers at its faster rate; for every 100 excimers it forms, its transfer on krypton moves 4.2 atoms to $1s_5$. On the kinetic EEDF the electrons sit at 2.55 eV, and the share differs from the two-temperature closure's by -5 %. The cycle's measured cross sections therefore decide the answer with the formation rates, with Kannari's fit in their place as a comparison [11, 12]:

5s → 5p cross sections	0 (no cycle)	× 0.6	× 1, Jung et al.	× 1.4	Kannari's fit
147 nm, kinetic EEDF	68.1	37.5	31.2	27.4	32.4
147 nm, two-temperature	69.4	39.5	32.9	28.8	33.6

Table 5: The 147-nm share (percent of E_{dep}) against the scale on the measured 5s → 5p cross sections, every level's. The band runs from 0.6 to 1.4, the measurements' quoted errors; the two $J = 1$ levels' proxies have a band of their own (`k_5s5p_j1`, Table 2).

Jung et al. measured the excitation out of the two metastable levels, $1s_5$ and $1s_3$, into the ten 5p levels, with stated uncertainties of ±35 and ±40 % [10]. For the two $J = 1$ levels nothing is measured; each takes the set of the metastable with the same ion core, shifted to its own threshold. At the reference point's 2.55 eV the four Maxwellian rates are 8.6×10^{-7}, 8.8×10^{-7}, 1.0×10^{-6} and 1.0×10^{-6} cm^3s^{-1} ($1s_5$, $1s_4$, $1s_3$, $1s_2$), and their weighted mean is ×1.16 Kannari's fit (×1.40 at 1 eV, ×1.23 at 2).

Pressure and power move the share as much (Table 6, two-temperature closure). It ranges from a fraction of a percent at 0.1 atm to 74 % at 2 atm and 100 kW cm^{-3}. At 0.1 atm a microsecond of this drive heats the gas to 2022 to 8549 K, beyond the measured range of several rates, where the excimer barely survives its redissociation.

kW cm^{-3}	0.1 atm	0.25 atm	0.5 atm	1 atm	2 atm
100	0.2	3.0	18.3	51.8	73.6
300	0.0	0.4	3.9	32.9	68.0
500	0.0	0.1	1.5	15.8	62.9

Table 6: The 147-nm share (percent of E_{dep}) over pressure and power, 1 μs pulses, the two-temperature closure on its field-driven tail table.

4 The two-temperature closure and its tail

The two-temperature closure keeps the electrons Maxwellian at T_e below krypton's lowest threshold, 9.9152 eV, and lets the distribution fall above it with a tail temperature $T_t = r\,T_e$, continuous at the break. It needs r, and takes it from a table. v0_1 to v0_3 read a table made for LIGHT's Kr/F_2 model from beam-driven states (its $F_2 = 0$ column), indexed by n_e/N alone, whose fast secondaries shape a tail otherwise than a field does. Along v0_4's own kinetic reference run that table gives 0.95 to 1.30 of the kinetic EEDF's tail ratio over the pulse body (1.13 on average), and over the scan's corners 0.67 to 1.68 of it (Table 7).

The field table. v0_4 builds its own, $r(n_e/N, T_e)$ on 238 points ($\log_{10} n_e/N$ from -9 to -2.5, T_e from 0.5 to 8 eV), from the model's own kinetic EEDF: ground-state krypton at 300 K and 1 atm, as many ions as electrons, a field at the mean energy $\frac{3}{2}T_e$, electron-electron collisions at n_e/N and the growth term, as the run solves them. At each point r is the tail ratio at which the two-temperature distribution removes the same energy through the 51 levels and the ionization as the kinetic one. At the reference point's state it gives 0.133, where v0_3's table gave 0.190. Where the kinetic tail is colder than the form can follow (low n_e/N and low T_e, 67 of the points) r takes the table's floor, 10^{-3}. Half and twice the pressure

move r by at most 3 % at the points the runs visit, through the Coulomb logarithm, and 700 K gas by 4 %.

What the table leaves out. Along the kinetic reference run the field table gives 0.82 to 0.85 of the kinetic EEDF's own tail ratio over the pulse body (0.84 on average; the kinetic ratio itself runs from 0.149 to 0.183). The table's states carry no excited atoms, and the run's 5s and 5p populations, through the electrons' 5s $\leftrightarrow$ 5p collisions, keep its tail hotter at the same mean energy; Penning electrons and recombination barely move it. Across the scan's corners its ratio to the kinetic tail runs from 0.73 to 1.24, below it at 2 atm and above it at 0.25. The closure's 147-nm share is nevertheless closer to the kinetic EEDF's on the field table at the reference point (+5 % against +8 %), and within 5 % of it at every corner, against 9 % on v0_3's table (Table 7): the Maxwellian below the break, which carries the 5s $\leftrightarrow$ 5p cycle, errs the other way, so at the reference point a tail as hot as the run's own would move the closure's share further from the kinetic EEDF's. The table is the right drive's, and its residual is documented rather than tuned away; a table with the excited populations as a third axis is the next step.

		147 nm (% of E_{dep})			r over the kinetic's		
atm	kW cm^{-3}	kinetic	field table	v0_3's table	field	v0_3's	T_g end (K)
0.25	100	2.85	2.99 (+5)	3.11 (+9)	1.24	1.68	1040
0.25	500	0.14	0.15 (+3)	0.15 (+3)	1.06	0.82	4027
2	100	72.96	73.63 (+1)	74.05 (+1)	0.73	1.20	328
2	500	61.22	62.88 (+3)	61.56 (+1)	0.79	0.67	491
1	300	31.22	32.92 (+5)	33.84 (+8)	0.84	1.13	707

Table 7: The two-temperature closure against the kinetic EEDF at the scan's corners and the reference point: the 147-nm share on the kinetic EEDF and on the closure with each tail table (its deviation in brackets), and each table's tail ratio over the kinetic run's own, averaged over the pulse body (50 ns to the pulse's end).

5 The electron swarm, and what the training did

The kinetic EEDF's solver is run as a swarm: no electron-electron collisions, the temporal growth term on, and every grid sized so that the EEDF falls 11 to 25 decades across it. It gives the drift velocity (the Joule power over eE), the ionization coefficient $\alpha/N = k_i/W$ and the characteristic energy D_T/μ, with the transverse diffusion taken over the same momentum transfer and growth term as the field's flux [8]. Swarm measurements are measurements of the cross sections themselves, so they may train them, while the device and beam measurements of Table 1 may not. v0_3 trained a few factors on Biagi's level-resolved cross sections against the swarm data that agree with one another, and v0_4 runs on them unchanged; this section is v0_3's, its numbers computed again on v0_4's code.

The data. 28 datasets, 441 points, from Dutton's survey [35] and LXCat's LAPLACE database [6, 7], which digitized them from the papers' tables or figures (Table 8). None prints an error, so each carries a class value: 3 % for a drift velocity measured after 1985, 5 % before, 7 % for a characteristic energy and 10 % for an ionization coefficient. A 2 % error in E/N enters through the model's own slope, which matters where the drift velocity climbs out of the Ramsauer minimum (0.01 to 0.03 Td) and for the ionization coefficient at low field. A dataset's χ^2 per point near 1 means the model agrees with it within these errors. Points above 300 Td are checks only: there the mean energy passes 7 eV, where the two-term approximation and the growth model part ways. The ionization coefficient below 50 Td never trains (below).

dataset	quantity	E/N (Td)	points	role	error (%)	χ^2 per point Biagi	χ^2 per point trained	rms (%) Biagi	rms (%) trained
Pack et al. 1962, 368 K [34, 35]	W	0.00153–0.00622	5	train	5	0.8	0.6	5	4
Hunter et al. 1988 [41]	W	0.002–3.0	42	train	3	0.8	0.7	3	3
Nakamura 1990 [43]	W	0.298–53	32	train	3	2.9	0.4	6	2
Pack et al. 1962, 300 K [34, 35]	W	0.000791–3.1	24	held out	5	4.5	4.3	12	11
Pack et al. 1992, 300 K [42]	W	0.0008–19	42	held out	3	14.2	17.7	15	15
Pack et al. 1962, 195 K [34, 35]	W	0.00121–0.0924	12	validation	5	3.1	2.8	10	9
Pack et al. 1992, 195 K [42]	W	0.0015–0.0894	10	validation	3	6.1	5.3	9	8
Bowe 1960 [45]	W	0.0247–7.4	53	held out	5	15.6	9.6	25	18
English and Hanna 1953 [46]	W	0.0399–2.0	31	held out	5	86.7	84.7	46	47
Dall'Armi et al. 1992 [44]	W	10–80	15	held out	3	10.3	13.4	11	12
Kucukarpaci and Lucas 1981 [47]	W	14–277	6	held out	5	39.6	48.4	25	26
the same, above 300 Td [47]	W	412–820	4	check	5	2.2	0.3	8	3
Koizumi et al. 1986 [48]	D_T/μ	0.005–0.9	26	validation	7	2.5	0.8	16	7
Al-Amin and Lucas 1987 [49]	D_T/μ	1.4–282	12	check	7	6.6	6.4	21	20
the same, above 300 Td [49]	D_T/μ	565–848	2	check	7	2.9	2.0	11	9
Kruithof 1940 [36, 35]	α/N	51–282	15	train	10	3.3	0.8	19	10
Heylen 1971 [37, 35]	α/N	55–250	21	train	10	0.9	0.2	10	5
Dall'Armi et al. 1992 [44]	α/N	55–80	6	train	10	1.8	0.8	15	11
Jacques et al. 1986 [5]	α/N	60–260	11	train	10	3.2	1.2	17	11
Bhattacharya 1979 [50]	α/N	62–281	11	train	10	0.4	1.3	8	15
Kruithof 1940, below 50 Td [36, 35]	α/N	14–45	11	held out	10	69.3	60.1	71	67
Specht et al. 1980, direct [51]	α/N	18–26	7	held out	10	15.0	10.4	53	46
Dall'Armi et al. 1992, below 50 Td [44]	α/N	20–50	7	held out	10	26.8	21.7	50	47
Heylen 1971, below 50 Td [37, 35]	α/N	28–49	10	held out	10	3.1	1.1	22	14
Jacques et al. 1986, 40 Td [5]	α/N	40–40	1	held out	10	0.0	1.2	2	17
Specht et al. 1980, total [51]	α/N	4.5–24	12	check	10	511	507	92	91
the same, above 300 Td [50]	α/N	312–625	4	check	10	1.1	2.7	12	19
the same, above 300 Td [36, 35]	α/N	339–1017	9	check	10	0.3	0.4	5	7

Table 8: Krypton's electron swarm data, their roles and the model's error on Biagi's file as published and on the trained set (v0_3's and v0_4's), each as the experiment measures it. Drift velocities W at 300 K unless labelled; Hunter et al.'s at 301 K. A validation set is held out of the training but took part in choosing which factors train (Table 10).

What a swarm experiment measures. The two-term solution with the temporal growth term gives the flux drift velocity and k_i/W. A drift tube times the swarm's centre of mass, the bulk drift velocity, and a steady-state Townsend experiment measures the spatial growth of its current, the SST coefficient. A Monte Carlo solution of the Boltzmann equation, with no two-term approximation, gives all of them [3]. Run on Biagi's file as published (Table 9), it finds the bulk drift velocity 23 % above the flux one at 300 Td, and the SST coefficient 9 % below the two-term's there. The two-term flux drift velocity matches the Monte Carlo's to 0.3 % from 10 to 300 Td, and from 40 Td up the two-term k_i/W lies 1.4 to 3.9 % below the Monte Carlo's flux coefficient. Below 20 Td, without ionization, the bulk and flux drift velocities are one, and the two-term drift velocity is compared as it is; its own error there (1.8 % at 0.2 Td) is absorbed by the trained factors, which are effective for the two-term solver. The two-term transverse diffusion is 46 to 79 % too high from 10 to 300 Td, so D_T/μ is scored only below 1 Td, and Al-Amin and Lucas's characteristic energies, 1.4 Td and up [49], are a check of the correction rather than of the cross sections. The model's two-term solution times these ratios is compared with each measurement. The ratios are taken on Biagi's file; on the trained set the Monte Carlo gives the drift velocity's and the ionization coefficient's to 1.6 %. Without them the model would be scored, and trained, against the wrong quantity: comparing the two-term coefficient with Townsend experiments puts it closer to them at high field than it is.

	Biagi's file			the trained set	
E/N (Td)	W_{bulk}	α_{SST}	D_T/μ	W_{bulk}	α_{SST}
20	0.999	1.036	0.63	–	–
30	1.003	1.036	0.60	1.002	1.034
40	1.006	1.036	0.58	–	–
50	1.011	1.025	0.57	1.018	1.008
75	1.038	0.997	0.56	–	–
100	1.065	0.979	0.56	1.070	0.973
150	1.113	0.956	0.59	–	–
200	1.152	0.939	0.62	1.165	0.930
300	1.232	0.906	0.68	1.241	0.908

Table 9: The Monte Carlo's value over the two-term solution's, LoKI-MC at 300 K and 1 Torr with 10^5 electrons [3]. The coefficient below 40 Td takes the 40-Td ratio, where the Monte Carlo counts too few ionizations to resolve it.

The training. The factors multiply Biagi's cross sections: the elastic momentum transfer by a factor read log-linearly between six energies (0.03 to 20 eV, held beyond them), and the excitation of the four 5s levels, the ten 5p levels and the 37 higher levels by one factor each. The factors at 0.03 and 0.2 eV stay at 1 (below), so 7 train. The ionization cross section is not trained: beam experiments measure it, and in a swarm its scale trades against the excitation's. The prior on each factor's logarithm is broad (standard deviation 0.5, a factor 1.65), so the data decide. Each trial of the factors needs the solver at every measured point, so the fits run on a neural network trained to reproduce it: 2000 sets of factors drawn over $\times 0.55$ to $\times 1.82$ (a Latin hypercube [53]), each solved at 81 values of E/N and gas temperature, and a three-layer network fitted to the change of each quantity from Biagi's file (Adam [55]). On designs it was not trained on it reproduces the solver to 0.10 % in the drift velocity, 0.13 % in D_T/μ and 0.74 % in the ionization coefficient, and every answer below is re-checked on the solver itself.

Which data train. The consistency check fits every candidate together, scores each against that joint fit, and drops the worst while its χ^2 per point exceeds 3. Of 16 candidates it drops 8: English and Hanna 1953 (73.7), Kucukarpaci and Lucas 1981 (44.6), Pack et al. 1992, 300 K (15.5), Dall'Armi et al. 1992 (16.4), Pack et al. 1992, 195 K (5.4), Pack et al. 1962, 300 K (5.0), Pack et al. 1962, 195 K (3.6), Bowe 1960 (3.5) (their χ^2 when dropped). The 8 that remain train. The choice of 3 matters little below it: at 2 the same sets remain. At 4, Pack et al. 1962, 195 K and Bowe 1960 join, and the 1960 drift velocities, -9 % from the 1988 ones where both exist on average, pull the momentum transfer at 6 eV to $\times 1.31$ and the 5s excitation to $\times 1.16$.

Which factors train. Two of the nine stay at 1, by a choice made on held-out data (Table 10). With all nine free the fit matches the train sets best (χ^2 0.49 per point pooled, against 0.64), but it lowers the momentum transfer at the Ramsauer minimum to $\times 0.88$ and raises it at 0.03 eV to $\times 1.13$, a trade the train sets allow and the data that weigh those energies differently reject: Koizumi et al.'s characteristic energy [48] goes from χ^2 2.5 per point on Biagi's file to 8.1, and Pack et al.'s drift velocities in gas at 195 K [34, 42] from 3.1 to 5.5 (1962) and from 6.1 to 8.7 (1992). With the factors at 0.03 and 0.2 eV held at 1, every train set stays within the consistency threshold (Hunter et al.'s drift velocity at 0.7 rather than 0.1), and those three fit better than on Biagi's file, at 0.8, 2.8 and 5.3, the best of the five variants. Holding the 0.6-eV factor too, or every momentum-transfer factor, fits both the train sets and these three worse. v0_3 trained that variant's 7 factors. Because these held-out sets chose it, they validate the training rather than test it. The tests that neither the training nor this choice saw are the ionization coefficient below 50 Td and the beam's partition (Section 7).

	train sets			held out (validation)		
factors trained	pooled	worst	Hunter 1988	Koizumi 1986	Pack 1962, 195 K	Pack 1992, 195 K
none (Biagi's file as published)	1.75	3.28	0.79	2.52	3.07	6.12
the three excitations	1.16	2.35	0.79	2.52	3.07	6.12
these, and the momentum transfer at 2 to 20 eV	0.66	1.26	0.75	2.43	2.89	5.29
these, and at 0.6 eV too (v0_3)	0.64	1.25	0.70	0.84	2.77	5.27
all nine	0.49	1.27	0.13	8.06	5.45	8.70

Table 10: Which factors train: each variant's least-squares factors on the train set, and the χ^2 per point on the solver itself, each dataset as its experiment measures it with Table 8's errors. Pooled is over the train sets' 143 points. v0_3 and v0_4 run on the fourth row's factors.

The fit. Least squares on the train set gives the factors; an ensemble MCMC [52] (acceptance 0.43) gives their posterior, and a neural posterior estimator, a network trained on simulated datasets to read the factors off them [54], gives the same answer independently (Table 11). Its $\pm 1\sigma$ intervals cover 0.60 to 0.65 of held-out simulations (ideal 0.68). The data pin the momentum transfer from 0.6 to 20 eV and the 5s and higher excitations (the network's R^2 at least 0.83); they barely pin the 5p excitation (R^2 0.15).

factor on	least squares	MCMC median	16–84 %	network	R^2
momentum transfer at 0.03 eV	1, not trained (Table 10)				
the same at 0.2 eV	1, not trained (Table 10)				
at 0.6 eV (the Ramsauer minimum)	1.157	1.157	[1.062, 1.260]	1.125 ± 0.09	0.93
at 2 eV	0.926	0.926	[0.891, 0.962]	0.909 ± 0.04	0.99
at 6 eV	1.039	1.040	[1.008, 1.071]	1.038 ± 0.03	0.99
at 20 eV and above	1.227	1.221	[1.106, 1.354]	1.240 ± 0.10	0.91
the four 5s levels (1s)	0.886	0.871	[0.776, 0.963]	0.838 ± 0.12	0.91
the ten 5p levels (2p)	0.925	0.954	[0.693, 1.293]	1.047 ± 0.34	0.15
the 37 higher levels	0.753	0.738	[0.631, 0.854]	0.703 ± 0.17	0.83

Table 11: The trained factors on Biagi's cross sections. v0_3 and v0_4 run on the least-squares values. The network column gives its mean and the standard deviation of the factor's logarithm; R^2 is the network's on held-out simulations.

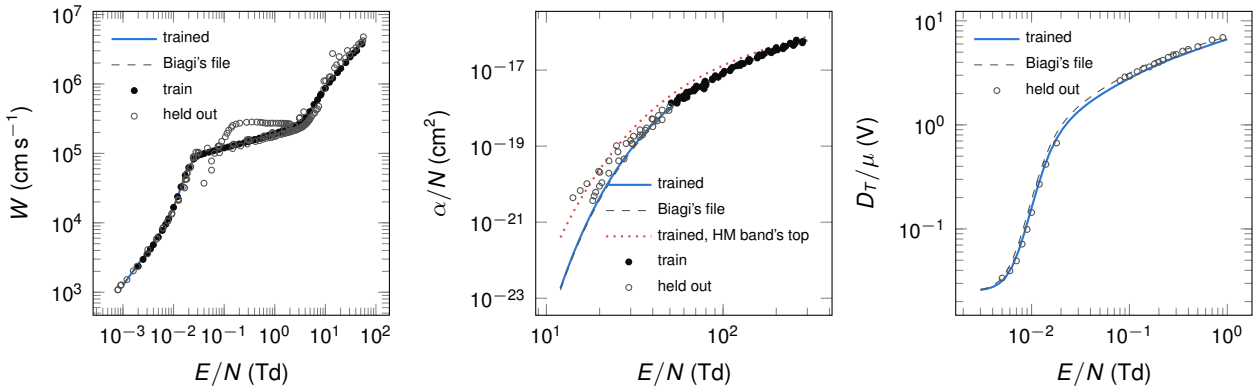


Figure 1: Krypton's drift velocity (left), first Townsend coefficient (middle) and characteristic energy (right) at 300 K, as the experiments measure them: the trained cross sections (solid), Biagi's file as published (dashed), and the measurements of Table 8, filled where they train and open where they are held out (the validation sets among them). Middle: the dotted curve is the trained set with every excitation of the levels at or above Kr_2^+'s energy ionizing associatively (the Hornbeck–Molnar band's top).

The scores.

- **The train sets.** The drift velocity of Hunter et al. [41], the most precise below 3 Td, moves from χ^2 0.8 to 0.7 per point, Nakamura's [43] from 2.9 to 0.4 (mean +5 to 0 %). The ionization coefficients of Jacques et al. [5] and Kruithof [36], which lay -15 and -18 % from the model, now lie -5 and -7 %; Bhattacharya's [50], which Biagi's file matched (-1 %), now lies +11 %. Every train set ends within the consistency threshold.
- **The held-out drift velocities** are the sets the check dropped. The training moves 5 of the 8 closer to the model, the two at 195 K among them, which took part in choosing which factors train. Three move further: Pack et al. 1992 at 300 K [42] (χ^2 14.2 to 17.7 per point), Dall'Armi et al.'s [44] (10.3 to 13.4) and Kucukarpaci and Lucas's [47] (39.6 to 48.4, -17 % from the model on average). Pack et al.'s 1992 set scatters by up to 30 % against the model between points within 5 % of each other in E/N; Bowe's [45] lie +16 % from the model, and English and Hanna's [46] -18 %.
- **The characteristic energy** of Koizumi et al. [48], held out, probes the momentum transfer near the Ramsauer minimum, which the drift velocities pin least: rms 15.8 % on Biagi's file and 7.1 % on the trained set. It chose which factors train, so it validates the training rather than tests it.
- **Below 50** Td, held out by rule, the mean error moves from -44 to -36 %: Heylen's from -20 to -9 %, Specht et al.'s direct coefficient [51] from -52 to -46 %, Kruithof's from -67 to -63 %.

Why the low field stays out. At 30 Td the electrons excite krypton 364 times for every ionization, so anything that ionizes an excited atom, or frees an electron at a surface with its light, can dominate the measured current. Specht et al. found the coefficient anomalously large at low field and attributed it to photoelectron emission at the surfaces by radiating metastables, which they say also explains why earlier measurements disagreed with transport calculations; their direct coefficient leaves that contribution out [51]. Bordage et al. found every published krypton set below the low-field data and ascribe it to Penning ionization of impurities [2]. And the 5 datasets part by up to $\times 4.0$ from one another where they overlap. Training on them would fit the cross sections to the purity and geometry of each experiment. Pure krypton has one process there of its own, the associative (Hornbeck–Molnar) ionization of atoms excited above 12.85 eV [4]. At the band's top it raises the trained set's coefficient $\times 12.7$ at 14 Td, $\times 3.1$ at 30 Td and $\times 1.5$ at 100 Td, and puts the model +54 % from the train sets on average, so a share that large cannot hold where those are measured. No krypton measurement gives the share, and it stays at 0 by default.

What this means for the reference point. The electrons sit at 1.34 to 17.5 Td. There direct ionization makes 0.13 % of the ions and Penning ionization the rest, so the low-field coefficient does not reach the answer. The excitation rates do, and the training moves them: on Biagi's file as published v0_4 gives 31.54 % at the reference point, and on the trained set 31.22 % (-1.0 %); the posterior of the factors spreads it by ± 0.11 points (Section 1).

6 The electroluminescence yield

Mano et al. measured the secondary scintillation of pure krypton at 1.1 bar and room temperature: the photons an electron drifting in a uniform field makes per centimetre, over the gas density [67]. Below the field at which the electrons ionize, every photon comes from an excitation, so the yield measures the excitation cross sections in the swarm regime, where no electron meets another and no excited atom meets an electron. Their fit, $Y/N = 0.113\,E/N - 0.3$ in 10^{-17} photons per electron per atom per cm^{-2} (E/N in Td), holds from its intercept at 2.7 Td to 14 Td. Nothing in v0_4 or in its training saw it.

v0_4's swarm (Section 5) gives the excitations per electron and per centimetre of drift, $\sum k_{\mathrm{exc}}/W$, level by level. Each excitation is then weighted by the photons the model's own network makes of it at 1.1 bar and 296 K: a dilute seed of each 5s level and of the 5p rung, with no drive and no electrons to speak of, run to the end of its cascade. At the bands' defaults every seed ends as one 147-nm photon (1.000 per $1s_5$ atom, 1.000 per $1s_4$ atom, 1.000 per 5p atom); with Kr_2^* + Kr at its band's top, 0.941 per 5p atom.

	Y/N (10^{-17} cm^2)		model over fit		
E/N (Td)	Mano et al.'s fit	model	bands' defaults	Kr_2^* + Kr at its top	5s share (%)
4	0.152	0.099	0.65	0.61	100.0
5	0.265	0.237	0.89	0.84	100.0
6	0.378	0.375	0.99	0.93	99.9
8	0.604	0.628	1.04	0.98	99.7
10	0.830	0.857	1.03	0.97	99.2
12	1.056	1.075	1.02	0.96	98.5
13.5	1.226	1.233	1.01	0.95	97.8

Table 12: The electroluminescence yield of pure krypton: Mano et al.'s fit [67] and v0_4's swarm at 296 K, each excitation weighted by its photons from the network's cascade at 1.1 bar; the 5s share is the excitations into the four 5s levels.

From 6 to 13.5 Td the model lies at 0.99 to 1.04 of the fit (2.4 % rms), and with the excimer's quenching at its band's top at 0.93 to 0.98. Toward the fit's intercept it falls below it (0.89 at 5 Td, 0.65 at 4): there the fit's yield is 0.22 and 0.12 of its value at 13.5 Td, and a straight line's intercept is the least constrained part of a fit to a yield that bends into its threshold. Nearly every excitation below 14 Td is a 5s level's, so the score tests the four 5s cross sections and the trained 5s factor ($\times$0.89) at the swarm's mean energies of 4.3 to 4.5 eV, and the excitation-to-photon chain at a density where nothing competes with the light.

7 Under an electron beam

Three measurements bear on the chain from excitation to the 147-nm photon. They were made under electron beams, so for them v0_4 runs under a beam drive. The drive is a fixture of this note, not part of the model. It deposits the given power as ion pairs, 5s and 5p atoms and secondary electrons in the proportions of a fast electron's degradation in pure krypton. That degradation is the Fowler–Spencer–Fano recursion on the same cross sections as the model, level by level, from 528 eV: an excitation continues the electron at $E - I$, and an ionization shares $E - I$ equally between the two electrons. Per ion pair it gives:

- W = 26.2 eV;
- 0.410 Kr**, from the levels above the 5s manifold;
- 0.273 5s atoms: 0.045 $1s_5$, 0.134 $1s_4$, 0.007 $1s_3$ and 0.088 $1s_2$, the two resonance levels most, as a fast electron's dipole excitation favours them (v0_3 put the four into one Kr*);
- 0.46 eV of heat, the higher levels' excess over the 5p rung's energy;
- 4.29 eV of sub-excitation secondaries, which the kinetic equation injects with a flat spectrum up to 9.9152 eV.

That is one excitation or ion per 15.55 eV.

The partition. Saito et al. [38] measured scintillation photons and electrons together under 5.49-MeV alpha particles in 1 to 10 atm of krypton. They found 0.55 excited atoms per ion pair, and W_s = 15.4 eV, the energy per photon when every pair recombines. They also confirmed that each recombination gives one photon. On the trained cross sections the drive's 15.55 eV per excitation or ion is +1.0 % from their W_s, its excitations per pair (0.68) differ from theirs by +24 %, and its W differs by +10 % from the 23.9 eV their two numbers imply. On Biagi's file as published the same degradation gives W = 27.6 eV (+16 %) and 0.79 excitations per pair (+44 %): the swarm data's smaller excitation moves both toward the measurement, which neither the training nor the choice of its factors saw. One photon of 8.43 eV per 15.4 eV is an efficiency of 0.547. The model's ceiling, one photon per excitation and per ion of the drive, is 0.542.

Low pump. Morozov et al. [39] measured the power a continuous 8 to 15-keV beam deposits in the gas against the VUV power emitted into 4π. For krypton they found 0.42 $\pm$ 0.05, flat from 500 to 1400 hPa

and over a tenfold range of beam power. At 1000 hPa, 30 and 300 W cm^{-3} for 1 μs with the light counted for 50 μs after, v0_4 gives 0.542 and 0.543 (Table 13). The error is +29 %, 2.5 of the measurement's standard errors (v0_3: +29 %). At this pump nothing competes with the light: 0.001 Penning ionizations per pair, every ion recombined (1.001 per pair), 0.000 excimers redissociated per pair, and every Kr_2^* radiating. So v0_4 sits at its ceiling, which Saito et al.'s measurement supports. Morozov et al. compare their value with calculated maxima of 0.46 and 0.52 for krypton. The 23 % between their value and Saito et al.'s one photon per excitation and per ion (0.547) is not explained in v0_4.

High pump. Eckstrom et al. [40] report the fluorescence efficiency of Kr_2^* near their theoretical limit of about 46 %. They found it under a low-current, long-pulse beam at modest pressures, for pump rates up to 3×10^5 W cm^{-3}. The abstract does not give the pressure of each point. At that pump rate (1 μs, the light counted for 20 μs) v0_4 gives:

- 0.131 at 0.5 atm, 0.466 at 1 atm and 0.543 at 2 atm, that is 76, 14 and 0 % below its ceiling of 0.542 (v0_3: 0.112, 0.342 and 0.523);
- at 1 atm, Penning ionization of the 5s levels on one another costs 0.243 photons per pair, because the excimer forms too slowly to outrun it entirely;
- at 1 atm with Kr* + Kr* at its band's low end, 0.514 (5 % below the ceiling), and with the $1s_5$ formation at its band's top, 0.472 (13 %).

Their limit assumes other numbers than v0_4's ceiling, so what the report tests is the loss: little at high pump. At 1 atm v0_4 now meets it (+1 %; v0_3: -26 %), losing 14 % below its ceiling where v0_3 lost more: its faster formation, out of $1s_4$ above all, outruns the Penning loss. At 0.5 atm the gas heats to 1318 K and the excimer redissociates (0.567 per pair), and at 2 atm v0_4 loses 0 %. The report gives no pressure for its points, so it no longer leans on the defaults.

run	measured	kinetic	two-temperature	v0_3, kinetic	Penning per pair	photons per pair	T_e (eV)	T_g end (K)
30 W cm^{-3}, 1000 hPa	0.42 ± 0.05	0.542	0.541		0.000	1.682	0.76	300
300 W cm^{-3}, 1000 hPa	0.42 ± 0.05	0.543	0.542	0.543	0.001	1.687	0.85	300
3×10^5 W cm^{-3}, 0.5 atm	near 0.46	0.131	0.130	0.112	1.278	0.407	0.38	1318
3×10^5 W cm^{-3}, 1 atm	near 0.46	0.466	0.464	0.342	0.243	1.448	0.54	615
3×10^5 W cm^{-3}, 2 atm	near 0.46	0.543	0.537	0.523	0.020	1.686	0.85	435
1 atm, Kr* + Kr* 1.5×10^{-10}		0.514	0.512		0.094	1.596	0.51	587
1 atm, $1s_5$ + 2Kr 4.2×10^{-32}		0.472	0.469		0.227	1.464	0.54	612

Table 13: v0_4 under the beam drive: the VUV efficiency (147-nm energy over the deposited energy) on both closures and v0_3's on the kinetic EEDF, and on the kinetic EEDF the Penning ionizations and 147-nm photons per ion pair, the pulse's mean T_e and the final gas temperature. The ceiling, one photon per excitation and per ion, is 1.683 photons per pair, an efficiency of 0.542; Saito et al.'s ideal is 0.547.

8 The energy ladder

Figure 2 draws v0_4's levels at their formation energies and the rows that move energy between them.

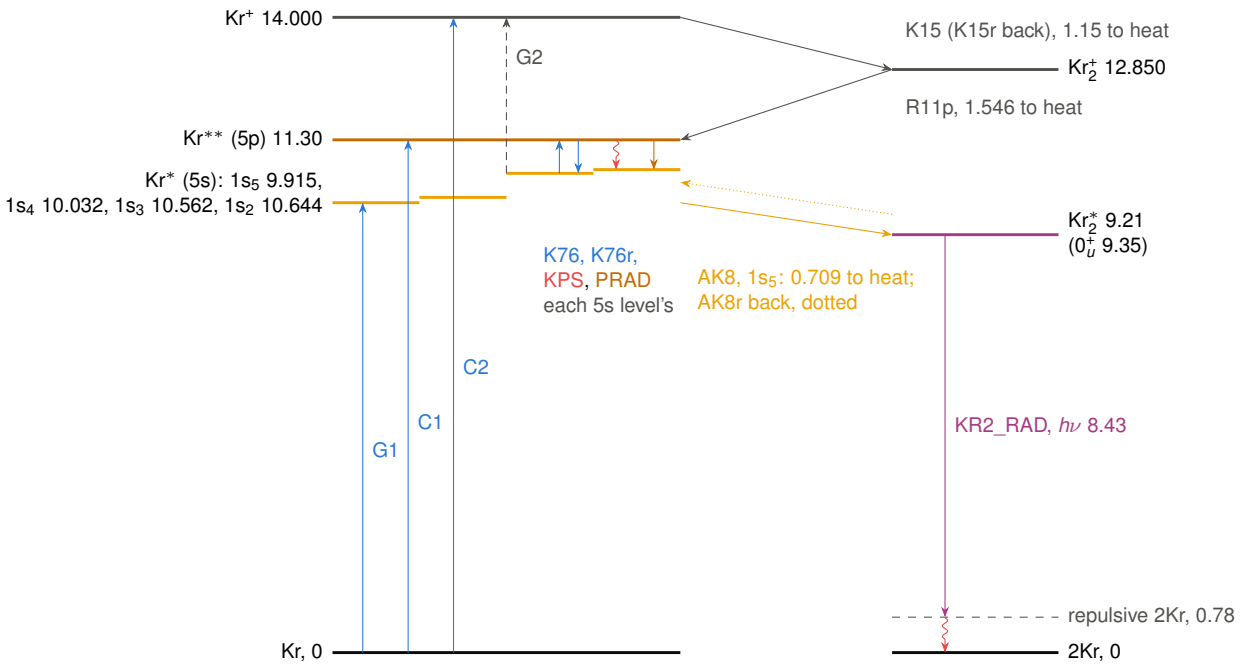


Figure 2: v0_4's energy ladder in eV above ground-state krypton at rest. A bar is a species at its formation energy; the four 5s levels are drawn side by side. Blue arrows are electron impact, which takes its energy from the electron gas; K76r, the 5p $\rightarrow$ 5s arrow beside K76, returns it. The dashed G2 is the Penning ionization of the 5s levels on one another, which gives the electron 5.831 eV for two $1s_5$ atoms. A red squiggle is heat: KPS relaxes the 5p rung on krypton, and PRAD radiates its gap to each level, 0.660 to 1.388 eV, in the near infrared. The excimer forms out of each 5s level, its well and the level's excess to heat, redissociates by detailed balance (AK8r), and radiates onto the repulsive ground state, which falls apart and heats the gas. The 5s levels exchange population on krypton (MIX, not drawn). C1's levels, 11.30 to 13.60 eV, cascade to the 5p rung, each heating the gas with its excess. HM, the Hornbeck–Molnar ionization of the levels above Kr_2^+'s energy, and the resonance lines of $1s_4$ and $1s_2$ are bands at 0 and not drawn.

9 Where v0_4 is not yet reliable

- **The excimer's formation.** Its rates still decide much of the answer. The $1s_5$ rate's measurements spread by a factor of 2.5; the $1s_4$ rate is one measurement [56], which Audouard et al.'s formation-time ratio puts three times lower [57] and Janssens et al.'s quasiclassical calculation 1.5 times higher [66]; the $1s_3$ rate is a proxy, the value of Kholin et al.'s measurement [68] not being in a source in hand. The temperature law is measured over 200 to 400 K [24] (v0_1 to v0_3 said 451 K, the range of a conversion), and the reference point ends at 707 K.
- **The field-driven regime is unscored.** No pure-krypton measurement in hand tests the 5s $\leftrightarrow$ 5p cycle under a field. Its cross sections are measured to $\pm$35 to 40 % out of the two metastables; those of the two $J = 1$ levels are not measured and take their metastable's (band `k_5s5p_j1`). The ledger books the 5p rung at the $2p_{10}$ energy, so an electron that pumps $1s_3$ or $1s_2$ pays that level's gap to it, 0.741 or 0.660 eV, where the core-preserving transitions out of those levels reach the $2p_1$ to $2p_4$ levels, 1.46 to 1.69 eV up, and their near-infrared lines carry the same gap.
- **The redissociation** rests on the excimer's well, whose measurements give 0.62 to 0.709 eV (its band), and on a rotational constant from an estimated bond length, 2.6 Å, the ion's, which enters as $1/r_e^2$; the ion's vibrational quantum is taken as the excimer's. At the reference point 5.8 % of the excimers formed redissociate; above 988 K at the fill's density most would.
- **Hot gas.** The 5p rung's relaxation by krypton carries no reverse, so the gas can de-excite the rung but not excite the 5s levels to it. By detailed balance that reverse out of $1s_2$ would match $1s_4$'s excimer

formation at the fill's density at 1213 K; below it, and at the reference point, it is negligible. The scan's 0.1 and 0.25 atm columns reach several thousand kelvin, where it is not, nor are the measured ranges of the formation and the recombination.

- **The excimer's two states.** The fast 0_u^+ state takes no share by default: its measured share is above 6×10^{19} cm^{-3} and appears from 1.5×10^{19} [57], so at 1 atm it lies between 0 and the band's top. Quenching by krypton is 0 by default, against one measurement of a pressure-dependent decay at 984 nm [64] that may be a transfer between radiating states rather than a loss. No rate mixes the two states, by krypton or by electrons.
- **The two-temperature closure** reads a tail table built from states without excited atoms. The run's 5s and 5p populations keep its tail hotter than the table's (Section 4), and its Maxwellian below the break errs the other way; the closure is +5 % from the kinetic EEDF at the reference point and within 5 % at the scan's corners. The default closure does not use it.
- **The trained cross sections** rest on errors the swarm data do not print (class values here) and on the consistency check's threshold (at 4 rather than 3, two more sets train and the momentum transfer at 6 eV moves to $\times 1.31$). The data barely pin the 5p excitation. Which factors train was chosen on held-out data (Table 10), so those sets validate the training rather than test it. Their posterior is propagated into the answer (± 0.11 points).
- **The rows with no measurement.** Kr_2^* + Kr_2^* and the stepwise ionizations of the 5s levels and of Kr** are 0 by default (Table 2). At its band's top Kr_2^* + Kr_2^* cuts the 147-nm share by 62 %, the stepwise ionization of the 5s levels by 47 % and that of Kr** by 24 %. The $J = 1$ levels' 5s $\rightarrow$ 5p cross sections are proxies, and their scale is the uncertainty's largest term.
- **Kr* + Kr*.** Its 5.0×10^{-10} cm^{3}s^{-1} is a 1974 measurement [18] as Kannari et al. print it [11], taken for every pair of levels.
- **The e-beam efficiency at low pump** is off Morozov et al.'s by +29 %, and v0_4 holds no process that would explain the difference.
- **The ionization coefficient below 50** Td lies -36 % from the measurements on average, which part by up to $\times 4.0$ from one another; every published cross-section set falls below them [2]. It is held out of the training by rule, and it does not reach the reference point, where Penning ionization makes the ions.
- **The Hornbeck–Molnar share** of the levels above 12.85 eV is not measured for krypton and is 0 by default. At its top it puts the model +54 % from the train sets and moves the reference point by -0.12 %.
- **Processes v0_4 does not carry**, each needing a measured rate before it can enter:
  - Kr_3^+ and its recombination;
  - electron collisions with Kr_2^*: ionization, quenching and the mixing of its two states;
  - Penning ionization between Kr* and Kr_2^*, and of Kr**;
  - the electrons' transfers between the 5s levels, and their superelastic return of the 5s levels to the ground state, which Blagoev et al. measured for thermal electrons in an afterglow [69] but which v0_4's kinetic EEDF would need as cross sections.

  Three-body recombination $Kr^+ + e^- + e^-$ [33] is not carried either: at 1 atm Kr^+ converts to Kr_2^+ in 7 ns.

How the numbers were made. Every number in this note is computed from the model when the note is built. None is typed by hand except the measurements, which carry their sources, and the training's record, which the model carries beside its cross sections and the training wrote. The runs are cached under a hash of the model's code, so a change to the model reruns them all (code `73e6a3962587`). The kinetic runs take about 8 minutes each; the two-temperature runs take seconds.

A Reaction rates

The rows of v0_4 with the values they run with and their sources, a `<level>` row once per 5s level. Electron-impact rows use their cross sections over the kinetic EEDF (or the two-temperature closure).

A band gives its default and its two ends.

row	reaction	value	source
G1_<level>	$e^- + Kr \rightarrow e^- + Kr^*(level)$	each 5s level at its threshold, 9.9152 to 10.644 eV, its own species	Biagi, Magboltz 8.97 [1, 6, 7], $\times 0.89$ (trained, Table 11)
C1	$e^- + Kr \rightarrow e^- + Kr^{**}$	the 5p, 4d, 6s and higher levels, 11.30 to 13.60 eV, $\times 0.92$ (5p) and $\times 0.75$ (higher) trained; the excess over 11.30 to heat	Biagi, Magboltz 8.97
HM	$e^- + Kr\ (+ Kr) \rightarrow Kr_2^+ + 2e^-$	the share of the excitations of the 20 levels at or above 12.85 eV; band 0 (default) to 1	no measurement for krypton; the process, Hornbeck and Molnar [4]
C2	$e^- + Kr \rightarrow Kr^+ + 2e^-$	13.9996 eV	Biagi, Magboltz 8.97
G6	e^- + Kr, elastic	the momentum transfer cross section, times the trained factor	Biagi, Magboltz 8.97
K76_<level>, K76r	$e^- + Kr^*(level) \rightleftarrows e^- + Kr^{**}$	the measured cross sections out of $1s_5$ and $1s_3$, the $J = 1$ levels on their same-core metastable's; band $\times$ 0.6 to 1.4, and $\times$ 0 to 1 on the $J = 1$ levels; the reverse by detailed balance over each level's gap	Jung et al. [10, 9]; Kannari's fit as a comparison [11, 12]
KPS_<level>	$Kr^{**} + Kr \rightarrow Kr^*(level) + Kr$	1.7×10^{-10} cm^{3}s^{-1} in all, over the levels by $2J + 1$ (band `kps_5s`: 1, to all into $1s_5$); band 3.0 $\times$ 10^{-11} to the default	Gazeli et al. [14], one level, products unresolved; the low end, the range Chang and Setser measured for argon's 4p levels [16]
PRAD_<level>	$Kr^{**} \rightarrow Kr^*(level) + h\nu$, near IR	τ = 29.0 ns, the $2J + 1$ mean of the ten 5p levels, branching by NIST's Einstein coefficients weighted by $2J + 1$	Dzierżęga et al. [17]; a TALIF lifetime of one level, 34.1 ns, as a cross-check [15]; NIST ASD [70]
G2_<a>_<b>	$Kr^*(a) + Kr^*(b) \rightarrow Kr^+ + Kr + e^-$	5.0×10^{-10} cm^{3}s^{-1} for each pair (twice for two levels); band 1.5×10^{-10} to the default	Leichner and Ericson [18] as Kannari et al. print it [11]
AK8_1s5	$Kr^*(1s_5) + 2Kr \rightarrow Kr_2^* + Kr$	3.8×10^{-32} cm^{6}s^{-1} $\times$ $(T_g/300)^{-0.64}$; band 1.7×10^{-32} to 4.2×10^{-32}	Marchal et al. [56]; Ueda et al. [19], Tracy and Oskam [21], Turner [59] and Smith and Turner [22] span the band; the law, Wieme and Lenaerts, 200 to 400 K [24]
AK8_1s4, AK8f	$Kr^*(1s_4) + 2Kr \rightarrow Kr_2^* + Kr$	9.2×10^{-32} cm^{6}s^{-1}, the same law; band to a third of the $1s_5$ rate; the share into the fast 0_u^+ state, band 0 to 0.17	Marchal et al. [56]; Audouard et al. [57]
AK8_1s3	$Kr^*(1s_3) + 2Kr \rightarrow Kr_2^* + Kr$	the $1s_5$ rate as a proxy; band 0 to the $1s_4$ rate	Kholin et al. measured it [68], the value not in a source in hand
AK8r_<level>	$Kr_2^* + Kr \rightarrow Kr^*(level) + 2Kr$	detailed balance on the excimer's well, D_0 0.709 eV; band to 0.62 eV	LaRocque et al. [60], Killeen and Eden [61]
MIX	$Kr^*(1s_4) + Kr \rightarrow Kr^*(1s_5) + Kr$, and $Kr^*(1s_2) + 2Kr \rightarrow Kr^*(1s_4) + 2Kr$	1.3×10^{-13} cm^{3}s^{-1} at 295 K on $(T_g/295)^{-1.29}$ through Turner's 196-K point, and 0.2×10^{-32} cm^{6}s^{-1}; their reverses by detailed balance	Turner [59]; Audouard et al. [58]
RES	$Kr^*(1s_4, 1s_2) \rightarrow Kr + h\nu$, 123.6 and 116.5 nm	the escaping share; band 0 to 1.7×10^5 s^{-1}	the pressure-independent term of Janssens et al.'s loss rate [66]
KR2_RAD	$Kr_2^* \rightarrow 2Kr + h\nu$, 147 nm	τ = 270 ns, $h\nu$ = 8.43 eV	Morikawa et al., 270 $\pm$ 5 ns [25]; 250 $\pm$ 20 [20], 280 $\pm$ 30 [23]
KR2F_RAD	$Kr_2^*(0_u^+) \rightarrow 2Kr + h\nu$	τ = 3.4 ns	Morikawa et al. [25]
QKR2	$Kr_2^* + Kr \rightarrow 3Kr$	0; band to 8.6×10^{-15} cm^{3}s^{-1}	Oka et al. [65]; Firestone and Selander [64]
PEN_KR2	$Kr_2^* + Kr_2^* \rightarrow Kr_2^+ + 2Kr + e^-$	0; band to 5.0×10^{-10}	no measurement; Kannari et al.'s value is the top [11]
K15, K15r	$Kr^+ + 2Kr \rightleftarrows Kr_2^+ + Kr$	2.4×10^{-31} cm^{6}s^{-1} $\times$ $(T_g/300)^{-0.75}$; the reverse by detailed balance on the ion's 1.149 eV	the mean of three measurements [26, 27, 28]; the law, Pérez-Ríos and Greene [29]; Signorell et al. [62]
R11p	$Kr_2^+ + e^- \rightarrow Kr^{**} + Kr$	1.6×10^{-6} $(300\,K/T_e)^{0.55}$ cm^{3}s^{-1}, $\pm$10 %; a gas-temperature factor, band 0 to 1	Shiu and Biondi [30]; Mikus within 8 % [31]; Cunningham and Hobson [63]
KSTEP_S, KSTEP_P	$e^- + Kr^*(level), Kr^{**} \rightarrow Kr^+ + 2e^-$	0; bands to 1.5×10^{-7} $T_e^{0.68} e^{-C/T_e}$ and 2.0×10^{-7} $T_e^{0.62} e^{-2.53/T_e}$ cm^{3}s^{-1}	no measurement; Vriens' calculation [13] as Kannari et al. print it [11]

Table 14: v0_4's rows. Rates in cm^3 s^{-1} or cm^6 s^{-1}, T_e in eV; C is 4.04 eV less the level's height above $1s_5$. Kr_2^+ lies 1.15 eV below Kr^+ + Kr [32].

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